

1_ Launch one EC2 using Amazon Linux 2 image and add a script in user data to install Apache.?

Open **AWS Console** → **EC2** → **Instances** → **Launch Instance**.

Enter a name, for example **AmazonLinux-Apache**.

Select **Amazon Linux 2 AMI**.

Select an instance type such as **t2.micro**/eligible free-tier type.

Select or create a key pair.

In **Network settings**, allow:

SSH – TCP 22

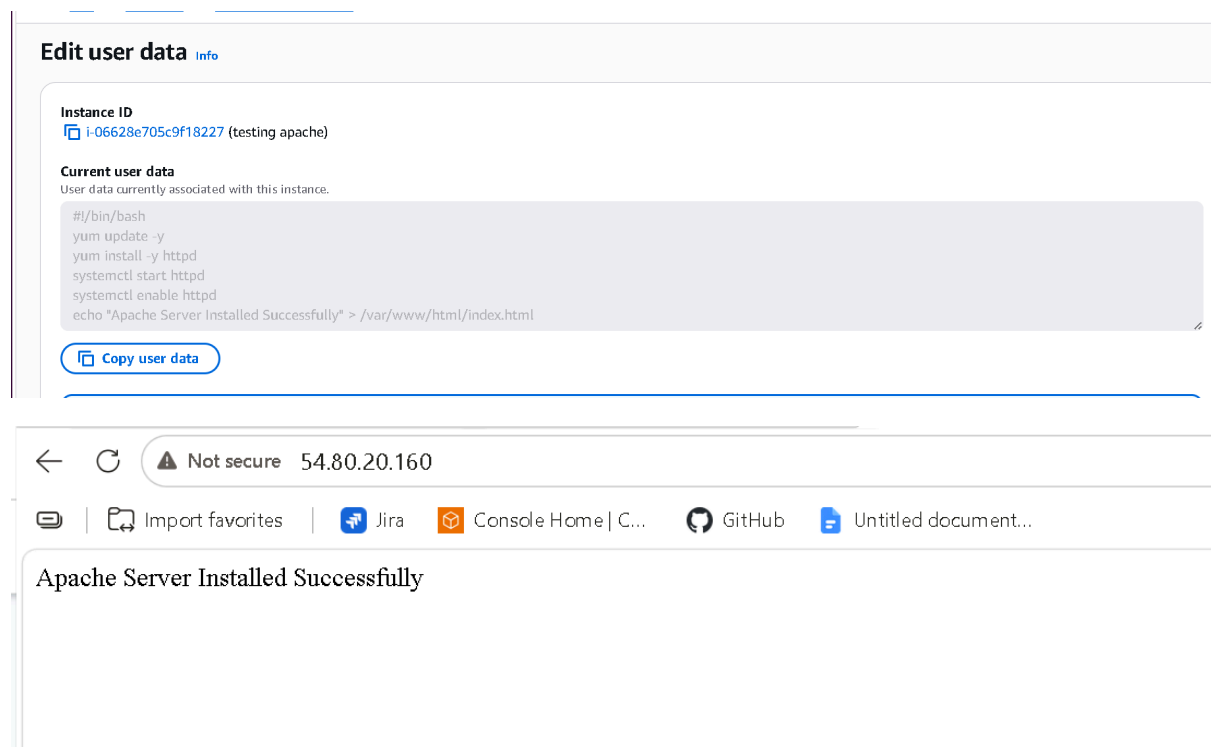
HTTP – TCP 80

Expand **Advanced details**.

Find **User data** and enter:

Click **Launch instance**.

After the instance is running, copy its **Public IPv4 address**



2 Launch one EC2 using Ubuntu image and add a script in user data to install Nginx.

Go to **EC2** → **Launch Instance**.

Enter a name such as **Ubuntu-Nginx**.

Select an **Ubuntu Server AMI**.

Select the required instance type.

Select/create a key pair.

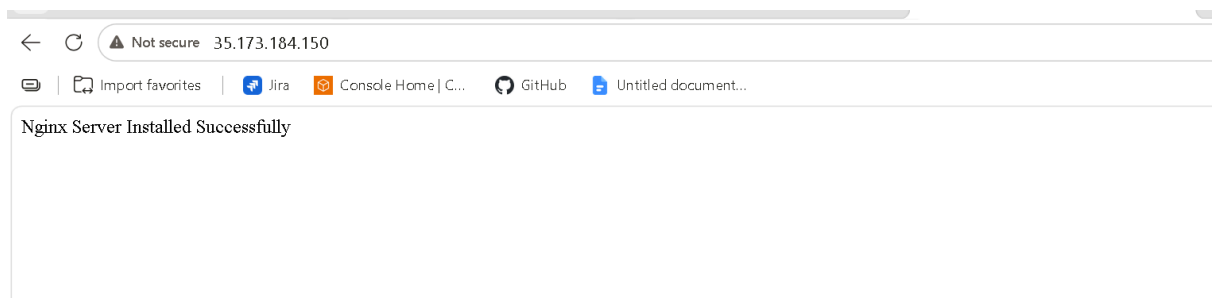
In **Security Group**, allow:

- SSH – TCP 22
- HTTP – TCP 80

Open **Advanced details** → **User data**.

Launch the instance.

Copy the **Public IPv4 address**



3 #Launch one Windows server and install Tomcat on Windows.

1. Go to **EC2** → **Launch Instance**.
2. Enter name:

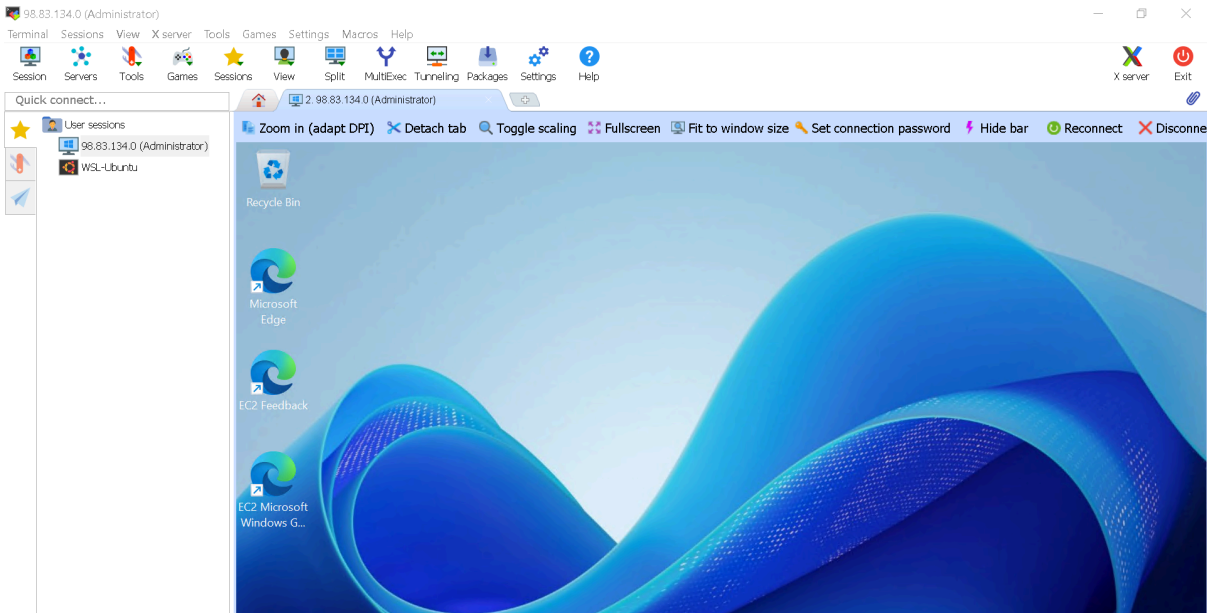
Windows-Tomcat

3. Select a **Windows Server AMI**.
4. Select an appropriate instance type.
5. Select/create a key pair.
6. Configure the Security Group:
 - RDP – TCP 3389
 - HTTP – TCP 8080
7. Click **Launch Instance**.

Connect to Windows

1. Select the running instance.
2. Click **Connect**.
3. Select **RDP client**.
4. Click **Get password**.
5. Upload your **.pem** key pair.
6. Click **Decrypt password**.
7. Copy the Administrator password.

- Download/open the RDP file.
- Connect using the Administrator account.



Description	File size	Download
ed Archive	210.6 MB	https://c
	188.75 MB	https://c
er	187.52 MB	https://c

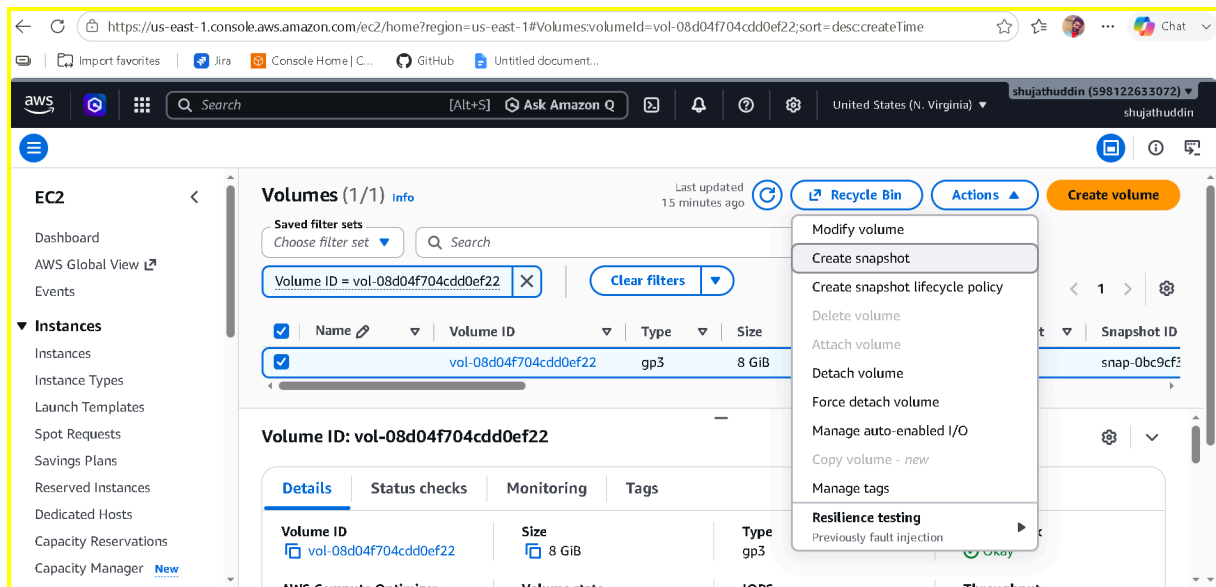
3 Take a snapshot of the instance created in Task 1.?

- Go to **EC2** → **Instances**.
- Select the Task 1 instance.
- Choose **Actions** → **Monitor and troubleshoot** → **Create snapshot** (the exact menu wording can vary in the current EC2 console).

4. Enter a description, for example:

Snapshot of Apache EC2

5. Click **Create snapshot**.
6. Go to **EC2** → **Snapshots**.
7. Verify that the snapshot is available.



Assign passwordless authentication for the EC2 created in Task 2.

1. Connect to the Amazon Linux 2 EC2 instance using MobaXterm.
2. Create a new user.
3. Generate an SSH key pair.
4. Copy the public SSH key.
5. Create the `.ssh` directory for the new user.
6. Add the public key to the `authorized_keys` file.
7. Set the correct permissions for the `.ssh` directory and `authorized_keys`.
8. Change ownership of the SSH files to the new user.
9. Exit from the EC2 instance.
10. Connect again using the new user and verify that no password is required.
11. Confirm the login using the `whoami` command.

Launch any EC2 using the spot purchasing option.

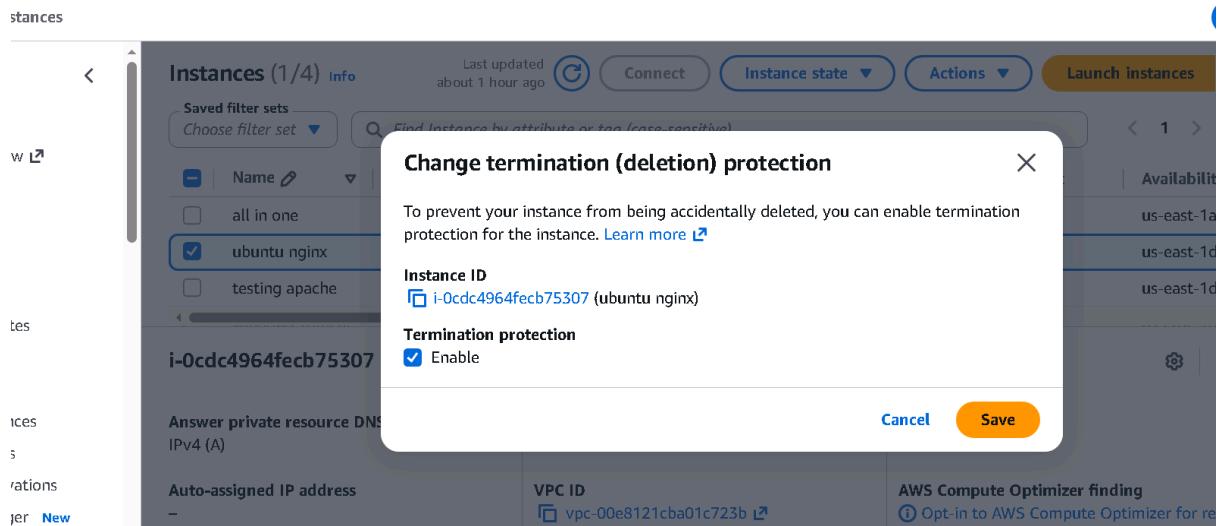
Enable termination policy on the EC2 created in Task 2.

1. Go to **EC2** → **Instances**.

2. Select the Ubuntu instance.
3. Choose:

Actions → Instance settings → Change termination protection

4. Enable **termination protection**.
5. Confirm the change.



Launch one EC2 using AWS CLI.

Open **Command Prompt** on the computer.

Configure AWS CLI using your AWS credentials.

Select the **AWS region** where the EC2 will be launched.

Find the required **AMI ID**.

Select an appropriate **instance type**, such as **t2.micro**.

Select an existing **key pair**.

Select a **security group**.

Select the required **subnet**.

Use the AWS CLI **run-instances** command to launch the EC2 instance.

Verify the instance status using the AWS CLI.

Open the **EC2 Console** and confirm that the new instance is in the **Running** state.

```
AWS Access Key ID [None]: arn:aws:iam::598122633072:user/wuhaib
AWS Secret Access Key [None]:
```

```
C:\Users\wuhaib>aws configure
```

```
Tip: You can deliver temporary credentials to the AWS CLI using your AWS Console session by running the command 'aws login'.
```

```
AWS Access Key ID [None]: AKIAYWQXDC5YN6PQEVHV
AWS Secret Access Key [None]: GY/sL7AS+Y3rgTdV/cAiUnNcjvsx1CK8/mRKJTfF
Default region name [None]: ap-south-1
Default output format [None]: json
```

```
C:\Users\wuhaib>aws sts get-caller-identity
```

```
{
  "UserId": "AIDAYWQXDC5YISIDPQW5L",
  "Account": "598122633072",
  "Arn": "arn:aws:iam::598122633072:user/wuhaib"
}
```

```
C:\Users\wuhaib>aws ec2 describe-images \
```

```
usage: aws [options] <command> <subcommand> [<subcommand> ...] [parameters]
To see help text, you can run:
```